1. Prime63 pilot launcher engineering record

TASK-20261017-4d8c00 · GOAL-ICPERF-e6b6a4 · 2026-10-17

This document reports engineering observations only. Full-width planted and archived natural ideals were not solved in this task.

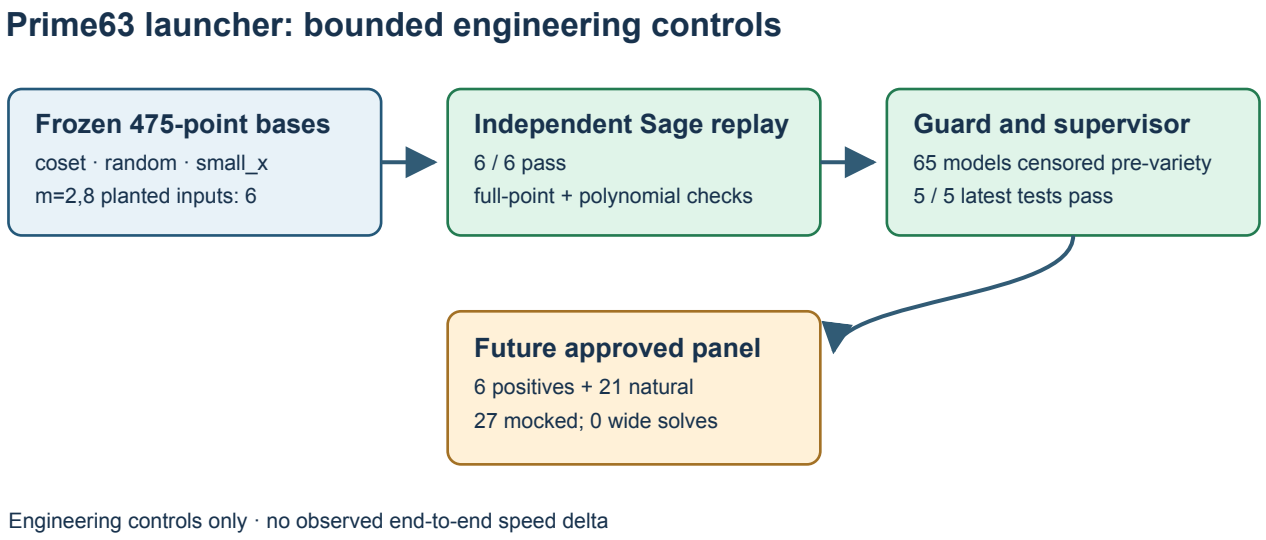


Figure 1: Preparation, independent replay, and bounded controls. The 21 natural systems are a future schedule.

1.1. Frozen inputs and checks

Six deterministic planted public inputs cover m=2 and m=8 for coset, random, and small_x, each with the original 475-point base. Signed selections and rejection streams are checker-only. Independent Sage replay passed all 6/6 full-point sums and polynomial-zero checks and rejected changed roots.

1.2. Guard and supervisor

The leading-ideal standard-monomial guard stops at 65 before variety enumeration. Both a 65-distinct-root and a repeated-root multiplicity-65 control returned model_cap_censored before variety. The repeated-root result is conservative censoring. A 64-standard-monomial control remained under cap.

The future 27-cell schedule applies 300 s per call, 8 GiB sampled process-tree RSS, 64 primary models, 4 h panel wall, and bounded SSD scratch/free space. Mocks accounted for 27 attempted / 0 unattempted and, on deliberate first failure, 1 attempted / 26 unattempted. Latest five tests passed.

An actual small-field worker returned two roots. In the latest pass, measured wall was 11.982 s; sampled peak tree RSS was 204,324,864 bytes; sampled peak scratch was 214,700 bytes; minimum SSD free was 540,807,798,784 bytes. A timeout control was terminated and reaped at 0.618 s.

1.3. Scope

Two failed engineering attempts and their tracebacks remain in the raw test log. Observed full-width solving degree, natural relation yield, useful rank, linear algebra, target discrete log, matched rho, and end-to-end delta are unmeasured. A later experiment requires a source snapshot, a complete frozen approved plan, positive replay, preapproval and a ready selector.

Exact inputs, commands, host state, and outcomes are bound in input_manifest.json, command.txt, environment.json, and engineering_validation.json beside this report.